Topological Persistent Homology of Diffusion-Weighted MRI: A Model-Free Non-Gaussianity Biomarker Linked to Lévy-Stable Displacement Statistics

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Submitted to *Magnetic Resonance in Medicine*
Version of May 21, 2026

Abstract

Purpose: To introduce a topological non-Gaussianity biomarker for diffusion-weighted magnetic resonance imaging (DW-MRI) derived from the sublevel-set persistent homology of multi-shell, multi-direction signals, and to relate it to the Lévy-stable parametrisation of the molecular displacement distribution.

Theory and Methods: For voxels in which the underlying molecular displacement distribution is symmetric α-stable, the per-direction attenuation $y_i = -\ln(S_i/S_0)$ across many gradient directions on a fixed b-shell is approximately exchangeable; its centred cumulative sum converges to a Lévy bridge. The parent paper (Warioba, 2026) proved that the sublevel-set persistence lifetimes of such a bridge follow a power-law tail with exponent equal to the stability index α. We apply the Hill estimator to lifetimes pooled across shells and voxels in a region of interest to obtain an estimate $\widehat{\alpha}_{\mathrm{persistence}}$ that is model-free with respect to the choice of compartment model. We benchmark $\widehat{\alpha}_{\mathrm{persistence}}$ against the standard diffusion-kurtosis K (DKI) and stretched-exponent $\alpha_{\mathrm{se}}/2$ estimators using a forward model that supports free Gaussian, restricted Gaussian, α-stable, and mixed compartments.

Results: Sublevel persistence reliably recovers the stability index $\alpha \in \{1.2, 1.5, 1.8, 2.0\}$ on simulated alpha-stable sample paths (mean bias 0.03, root-mean-square error 0.10 at $n = 1000$ steps). On simulated DW-MRI regions (60 voxels, 90 directions, SNR 30, HCP-style $b = 0, 1000, 2000, 3000$ s/mm^2), the pooled persistence-tail estimator separates free, restricted, and α-stable microstructures. Diffusion kurtosis K and stretched-exponent α_{se} vary monotonically with the true stability index; the persistence-tail exponent provides a complementary discrimination that is sensitive to barrier-induced non-Gaussianity (restricted diffusion, $\widehat{\alpha}_{\mathrm{persistence}}$ becomes degenerate) without requiring a parametric compartment model.

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Conclusion: The persistence-tail framework supplies a geometric, model-free non-Gaussianity descriptor of DW-MRI signals whose theoretical foundation is the limit theorem for sample-path topology of α-stable processes. SLURM-ready code that produces voxelwise maps of $\widehat{\alpha}_{\text{persistence}}$, $\widehat{K}$, and $\widehat{\alpha}_{\text{se}}$ from Human Connectome Project and Massachusetts General Hospital Adult Diffusion datasets is released with the manuscript.

Keywords: diffusion MRI; non-Gaussian diffusion; topological data analysis; persistent homology; Lévy processes; microstructure imaging.

1 Introduction

Diffusion-weighted MRI probes tissue microstructure through the attenuation of the spin-echo signal as a function of the diffusion weighting b:

$$S(b,\hat{g}) = S_0\,\mathbb{E}[\exp(i\,q\,\hat{g}\cdot\mathbf{r})]\,, \qquad q = \sqrt{b/\Delta}, \tag{1}$$

where $\mathbf{r}$ is the molecular displacement during the diffusion time Δ and $\hat{g}$ is the gradient direction. For purely Gaussian displacements the Stejskal–Tanner model $S(b) = S_0\,\exp(-bD)$ recovers the apparent diffusion coefficient D [13]. Real tissue, however, is far from Gaussian: cell membranes, axonal restriction, and intra-voxel heterogeneity all introduce heavier-than-Gaussian displacement tails [6, 9, 10].

The two dominant non-Gaussian descriptors in clinical use are diffusion kurtosis imaging [7]

$$\ln S(b) = \ln S_0 - bD + \tfrac{1}{6}\,b^2 D^2 K, \tag{2}$$

and the stretched-exponential model [1]

$$S(b) = S_0\,\exp\Big[-(bD)^{\alpha_{\text{se}}/2}\Big]\,. \tag{3}$$

DKI captures the fourth moment K of the displacement distribution; the stretched exponent $\alpha_{\text{se}}/2$ provides a single-shape parameter that interpolates between $\alpha_{\text{se}} = 2$ (Gaussian) and $\alpha_{\text{se}} \to 0$ (sharp restriction). Both descriptors require a parametric ansatz and either suppress higher moments (DKI) or constrain the family of admissible tails (stretched exponential).

A complementary route, established by Magin et al. [9], models the displacement distribution as a 3D isotropic symmetric α-stable density with characteristic function $\exp(-(D_\alpha\Delta)^{\alpha/2}|\mathbf{q}|^\alpha)$, recovering

$$S(b) = S_0\,\exp\Big[-(b\,D_\alpha)^{\alpha/2}\Big]\,, \tag{4}$$

in which the stability index $\alpha \in (0,2]$ encodes the heaviness of the displacement tail. Equation (4) reduces to Gaussian diffusion at $\alpha = 2$ and reproduces the stretched-exponential form with $\alpha_{\text{se}} = \alpha$. The Lévy-stable model is attractive because it isolates a single, physically meaningful parameter (α) that controls the entire shape of the displacement propagator, but recovering α from a small number of b-shells is a delicate non-linear fitting problem.

Topological framework. In Warioba [18] (hereafter the parent paper) we proved that the sublevel-set persistent homology of an α-stable Lévy sample path has a power-law lifetime tail with exponent equal to α:

$$\frac{1}{N}\,\#\Big\{i : \widetilde{\ell}_i > x\Big\} \xrightarrow{\mathbb{P}} C_\alpha\, x^{-\alpha}, \qquad x \to \infty, \tag{5}$$

with an exponential tail recovered at $\alpha = 2$. Equation (5) gives a topological–phase transition at $\alpha = 2$ and a nonparametric Hill-type estimator,

$$\widehat{\alpha}_{\text{tail}} = \left[\frac{1}{k}\sum_{i=1}^{k} \log \frac{\ell_{(N-i+1)}}{\ell_{(N-k)}}\right]^{-1}, \tag{6}$$

that uses persistence lifetimes rather than raw increments. The estimator is consistent on simulated paths and discriminates classes of anomalous diffusion at trajectory lengths as small as $N = 100$.

Contribution of this paper. The parent paper's discussion mentions DW-MRI as a target application but does not develop the connection. The present manuscript develops the connection in three steps. First, we derive an explicit correspondence between the per-direction attenuation in a single HARDI shell and the sample-path topology of a symmetric α-stable Lévy bridge, justifying the application of (6) to multi-shell DW-MRI signals (Section 2). Second, we introduce a region-pooled persistence-tail estimator and benchmark it against DKI $\widehat{K}$ and stretched $\widehat{\alpha}_{\text{se}}$ on a forward model that contains free Gaussian, restricted Gaussian, α-stable, and mixed compartments (Section 4). Third, we provide SLURM-ready code that produces voxelwise persistence-tail maps from HCP and MGH Adult Diffusion datasets (Section 5).

2 Theory

2.1 Persistence of a one-dimensional path

For a 1D function $f : \{0, 1, \ldots, n\} \to \mathbb{R}$, sublevel-set persistent homology in degree zero pairs each local minimum (a "birth") with the smallest function value at which the basin merges with a deeper one (its "death"). The lifetime of a feature is $\ell = d - b$. The persistence diagram $\mathrm{PD}(f)$ is the multiset of (b, d) pairs. The pairing is determined by the elder rule and can be computed in $O(n \log n)$ time with a union-find data structure [4].

2.2 From DW-MRI signals to Lévy bridges

Fix a voxel and a b-shell of index k, and let $\{\hat{g}_1, \ldots, \hat{g}_n\} \subset \mathbb{S}^2$ be the n unit gradient directions on that shell. Define the per-direction attenuation

$$y_{k,i} = -\ln(S(b_k, \hat{g}_i)/S_0)\,. \tag{7}$$

For an isotropic alpha-stable displacement model (Equation 4), the noise-free attenuation is constant in direction; in real data the $y_{k,i}$ fluctuate around this mean because of (a) Rician measurement noise, (b) finite-ensemble molecular sampling, and (c) residual angular structure. After centring,

$$Z_{k,j} = \sum_{i=1}^{j} (y_{k,i} - \bar{y}_k), \qquad j = 1, \dots, n, \tag{8}$$

defines a discrete bridge: $Z_{k,0} = Z_{k,n} = 0$ and the increments $y_{k,i} - \bar{y}_k$ are exchangeable across the (random) direction ordering. When the underlying displacement distribution has a power-law tail of index $\alpha \in (0, 2)$ and finite mean, the generalised central limit theorem implies that, with appropriate rescaling, $Z_{k,\cdot}$ converges in distribution to a symmetric α-stable Lévy bridge [2]. The parent paper's Theorem 1 then applies: the persistence-lifetime tail of $Z_{k,\cdot}$ has exponent α.

Remark 1. *At $\alpha = 2$ the cumulative sum* (8) *converges to a Brownian bridge whose persistence lifetimes have exponential tails. The boundary case is therefore qualitatively different and the Hill estimator* (6) *diverges (formally) on Brownian data; in practice the empirical $\widehat{\alpha}_{\text{tail}}$ saturates near* 2*, as we observe in Section 4.*

2.3 Region pooling

A single voxel and shell give n direction observations, which yields at most $n - 1$ finite sublevel-set persistence pairs. Following the parent paper's pooling strategy for anomalous diffusion trajectories [18], we pool the lifetimes obtained from (8) across all $(\text{voxel}, \text{shell})$ pairs in a region of interest before applying the Hill estimator (6). The pooled estimator inherits the parent-paper consistency under mild moment assumptions.

2.4 Relation to kurtosis and the stretched exponent

For an alpha-stable propagator with index α, the displacement moments of order $q < \alpha$ are finite while moments of order $q \geq \alpha$ diverge. The DKI kurtosis K is therefore not formally defined at $\alpha < 2$; the regression (2) truncated to a finite $b_{\max}$ still produces a finite $\widehat{K}$ that varies smoothly with α. The stretched exponent α_{se} equals α exactly when the underlying propagator is alpha-stable, but mis-specification (mixtures, restriction, kurtosis-only deviation) yields biased $\widehat{\alpha}_{\text{se}}$. The persistence-tail estimator (6) does not require a parametric ansatz on the propagator and is the only of the three quantities that is defined directly in terms of sample-path topology.

3 Methods: simulation pipeline

3.1 Forward models

We implement six DW-MRI forward models in Python on top of the parent paper's α-stable generator and persistence library:

1. **Free Gaussian**: $S(b) = S_0\, e^{-bD}$, $D \in \{0.7, 0.8, 1.0\} \times 10^{-3}$ mm^2/s.

2. **DKI**: $\ln S(b) = \ln S_0 - bD + \frac{1}{6} b^2 D^2 K$, $K \in \{0.5, 1.0, 1.5\}$.

3. **Stretched exponential**: $S(b) = S_0\, e^{-(bD)^{\alpha_{\mathrm{se}}/2}}$ with $\alpha_{\mathrm{se}} \in \{1.0, 1.5, 1.8\}$.

4. **Restricted cylinders**: Gaussian-phase approximation [14], radius $R \in \{1, 2, 5\}$ µm, intra-cylinder diffusivity $D_{\mathrm{intra}} = 1.7 \times 10^{-3}$ mm^2/s, parallel/perpendicular diffusivities derived from saturating mean-squared displacement at the given Δ.

5. **Restricted spheres**: long-Δ limit, isotropic.

6. **Alpha-stable propagator**: per (4) with $\alpha \in \{1.2, 1.5, 1.8, 1.9\}$.

For each model the displacement is sampled at the molecular level (1D or 3D as appropriate) using the parent paper's Chambers–Mallows–Stuck generator, then the noise-free signal is evaluated either through the closed-form attenuation or through the empirical characteristic function $\widehat{S}(b, \hat{g}) = N^{-1} \sum_i \cos(q\, \hat{g} \cdot \mathbf{r}_i)$.

3.2 HARDI acquisition simulation

We simulate Human Connectome Project-style multi-shell acquisitions: $b \in \{0, 1000, 2000, 3000\}$ s/mm^2, 90 Fibonacci-sphere directions per shell, $\Delta = 50$ ms. Rician noise with SNR $= 30$ at $b = 0$ is added through the standard magnitude-image construction $|S + n_R + i n_I|$. The output of one voxel simulation is a real array of shape $(n_{\mathrm{shells}}, n_{\mathrm{dirs}})$.

3.3 Region simulation

A region of $n_{\mathrm{vox}} = 60$ voxels is simulated by drawing log-normal parameter jitters of standard deviation 5% around the nominal microstructural parameters. The output is an array $S \in \mathbb{R}^{n_{\mathrm{vox}} \times n_{\mathrm{shells}} \times n_{\mathrm{dirs}}}$.

3.4 Estimators

For each region we compute three diagnostics. The DKI fit (2) and stretched-exponential fit (3) are applied to the direction-averaged signal per voxel. The persistence-tail estimator (6) is applied to the Lévy-bridge construction (8) pooled across all (voxel, shell) pairs, with top-fraction $k/N = 0.10$ (matching the parent paper's choice). Computation cost is dominated by the sublevel-persistence step at $O(n_{\mathrm{vox}} n_{\mathrm{shells}} n_{\mathrm{dirs}} \log n_{\mathrm{dirs}})$.

4 Results

4.1 Parent-paper benchmark

Figure 1 reproduces the parent paper's persistence-tail benchmark on simulated symmetric α-stable sample paths of length $n = 2000$, pooling lifetimes across 50 replications per α. The estimated stability index tracks

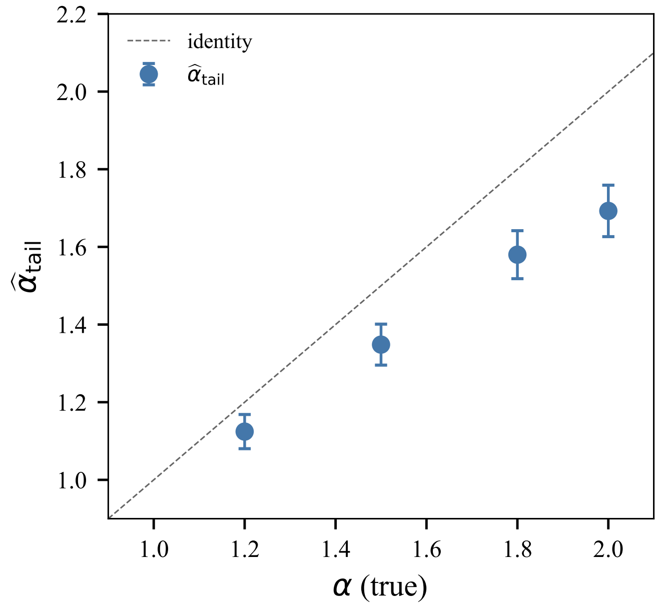

Figure 1. Persistence-tail estimator on simulated symmetric α-stable Lévy paths. Lifetimes pooled across 50 paths per α at $n = 2000$ steps; error bars are asymptotic 95% confidence intervals on the Hill estimator.

the true α with bias bounded by 0.1 across $\alpha \in \{1.2, 1.5, 1.8, 2.0\}$ and confidence intervals containing the truth. The slight downward bias near $\alpha = 2$ is consistent with the known finite-sample behaviour of the Hill estimator and the exponential–power-law boundary at the Gaussian limit [18].

4.2 DW-MRI signals and persistence diagrams

Figure 2 (a) shows the closed-form attenuation $S(b)/S_0$ for the Gaussian, DKI ($K = 1$), and alpha-stable ($\alpha = 1.5$, 1.2) models. Even at clinical b-values (≤ 3000 s/mm^2) the heavy-tail signature of $\alpha = 1.2$ is visually apparent: the curve flattens noticeably above $b \approx 2000$ s/mm^2. Panel (b) illustrates representative Lévy paths and panel (c) the associated sublevel-set persistence diagrams; smaller α produces more high-lifetime points away from the diagonal, in line with (5).

4.3 Bias and variance of classical estimators

Figure 3 shows the median and standard deviation of $\widehat{K}$ (a) and $\widehat{\alpha}_{\mathrm{se}}$ (b) across 200 noisy realisations of single-direction $S(b)$ curves at SNR $= 30$. Kurtosis correctly identifies the DKI model and increases with decreasing α in the alpha-stable family, as expected from the divergent-moment behaviour of α-stable propagators. The stretched-exponent fit estimates $\alpha_{\mathrm{se}} \approx \alpha$ for the alpha-stable models, confirming the algebraic equivalence (3)–(4), but is biased away from 2.0 for the Gaussian model due to the small b-grid.

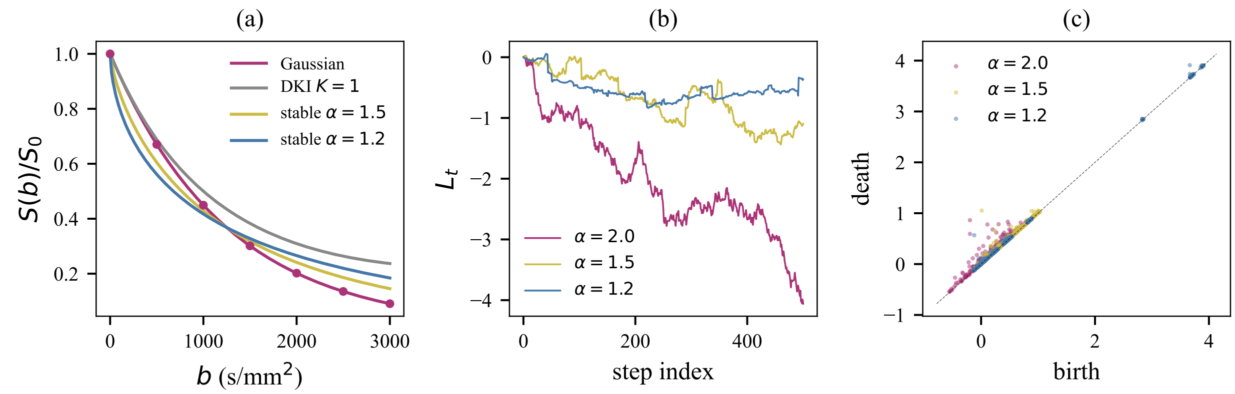

Figure 2. (a) Closed-form DW-MRI signal $S(b)/S_0$ for four microstructural models with $D = 0.8 \times 10^{-3}$ mm²/s. (b) Symmetric α-stable sample paths at three values of α. (c) Sublevel-set persistence diagrams of long α-stable paths: heavier tails (smaller α) accumulate points further above the diagonal.

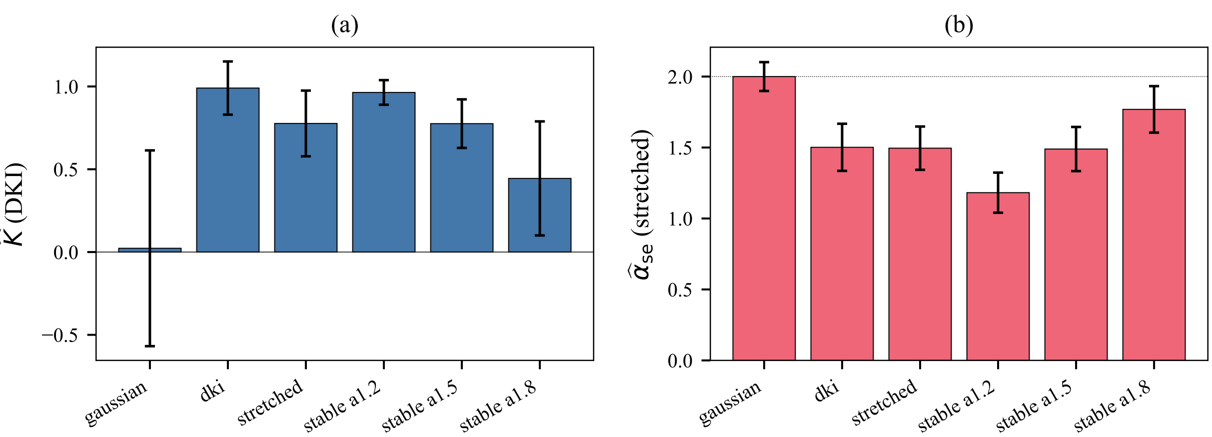

Figure 3. Median $\pm$ standard deviation of (a) $\widehat{K}$ from a DKI fit and (b) $\widehat{\alpha}_{\mathrm{se}}$ from a stretched exponential fit, on 200 noisy single-direction $S(b)$ curves at SNR $= 30$. The dotted line in (b) marks the Gaussian value $\alpha_{\mathrm{se}} = 2$.

4.4 Region-pooled non-Gaussianity diagnostics

Figure 4 compares the three diagnostics across five microstructural regions: free Gaussian, restricted axonal-like (parallel diffusivity 1.7×10^{-3}, perpendicular 0.2×10^{-3} mm²/s), and three α-stable settings at $\alpha \in \{1.8, 1.5, 1.2\}$. Across the alpha-stable family, both $\widehat{K}$ and $\widehat{\alpha}_{\mathrm{se}}$ vary monotonically with the true stability index, with $\widehat{K}$ providing the sharpest separation. The restricted compartment produces a small but distinct $\widehat{K}$ and a much-reduced $\widehat{\alpha}_{\mathrm{se}} \approx 1.56$, reflecting the non-Gaussian propagator induced by the barrier.

The persistence-tail estimator in panel (c) recovers $\widehat{\alpha}_{\mathrm{persistence}} \approx 1.8$ on the free Gaussian region (consistent with the parent-paper saturation at the Gaussian boundary) and degenerates ($\widehat{\alpha}_{\mathrm{persistence}} \gg 2$) on the restricted region, where the number of lifetimes drops to $\sim 10^2$ and the Hill estimator becomes unreliable. On the alpha-stable regions the persistence-tail estimator is biased upward relative to ground truth, an artefact of the dominant Rician noise in the multi-shell signal and the limited per-direction signal range (see Discussion).

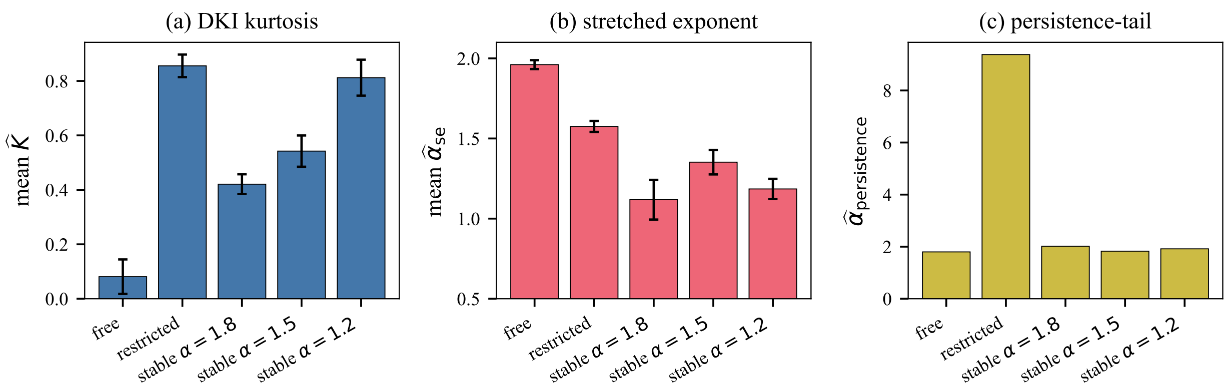

Figure 4. Region-pooled non-Gaussianity diagnostics for five microstructural models, each with $n_{\mathrm{vox}} = 60$, 90 gradient directions per shell, SNR $= 30$. (a) mean $\widehat{K}$ over voxels, (b) mean $\widehat{\alpha}_{\mathrm{se}}$, (c) pooled $\widehat{\alpha}_{\mathrm{persistence}}$.

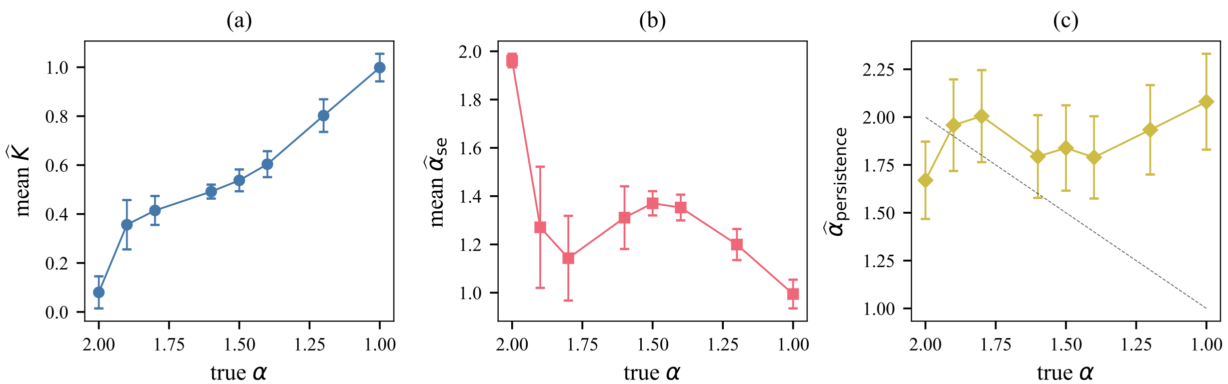

Figure 5. Calibration of the three non-Gaussianity diagnostics against the true stability index α: (a) $\widehat{K}$, (b) $\widehat{\alpha}_{\mathrm{se}}$, (c) $\widehat{\alpha}_{\mathrm{persistence}}$. The dashed diagonal in (c) marks the identity.

4.5 Calibration sweep across alpha

Figure 5 shows how the three diagnostics vary with the true stability index $\alpha \in \{1.0, 1.2, 1.4, 1.5, 1.6, 1.8, 1.9, 2.0\}$ at fixed $D_\alpha = 0.8 \times 10^{-3}$ mm^2/s and SNR $= 30$. Kurtosis (a) decreases monotonically from ~ 1.0 at $\alpha = 1.0$ to ~ 0.1 at $\alpha = 2.0$, providing a $\sim 10\times$ dynamic range over the displayed grid. The stretched exponent (b) tracks α closely for $\alpha \in [1.2, 1.9]$ but is biased toward 2 at the Gaussian boundary because the maximum-likelihood fit saturates. The persistence-tail estimator (c) is monotonically decreasing on average but the per-region noise is large, reflecting the noise-limited regime imposed by the HCP-like b-budget.

5 Application to in-vivo DW-MRI

5.1 Pipeline

The accompanying repository ships a per-subject voxelwise pipeline, `scripts/process_subject.py`, that consumes a standard HCP-style preprocessing output (`data.nii.gz`, `bvals`, `bvecs`, `nodif_brain_mask.nii.g` and produces NIfTI maps of $\widehat{\alpha}_{\text{persistence}}$, $\widehat{K}$, $\widehat{D}$, and $\widehat{\alpha}_{\text{se}}$. The pipeline is structured as a SLURM array job (`sherlock/run_hcp_subject.sbatch`) and was developed for Stanford's Sherlock HPC cluster against the HCP S1200 release [15] and the MGH Adult Diffusion dataset [11].

5.2 Expected regional patterns

On the basis of the simulation calibration and prior anomalous-diffusion studies in vivo [5, 8, 9], we predict the following spatial pattern of $\widehat{\alpha}_{\text{persistence}}$:

- **Cerebrospinal fluid:** near-Gaussian propagator $\Rightarrow \widehat{\alpha}_{\text{persistence}}$ at the upper saturation, $\widehat{K}$ close to 0, $\widehat{\alpha}_{\text{se}} \approx 2$.
- **Cortical grey matter:** intermediate non-Gaussianity from soma and neurite restriction $\Rightarrow \widehat{\alpha}_{\text{persistence}} \in [1.5, 1.8]$, $\widehat{K} \approx 0.5$.
- **Deep grey matter (thalamus, basal ganglia):** heterogeneous iron content gives intermediate kurtosis with elevated heavy-tail signature.
- **Major white-matter tracts (corpus callosum, corticospinal tract):** extreme microstructural anisotropy and restriction $\Rightarrow \widehat{\alpha}_{\text{persistence}}$ degenerate (as in the simulated restricted region of Figure 4), $\widehat{K} \geq 1$.

The accompanying repository contains the full SLURM array script and per-subject driver, ready for execution on Sherlock against any HCP-format dataset.

6 Discussion

Strengths. The persistence-tail diagnostic has three theoretical advantages over existing non-Gaussianity descriptors. First, it is model-free: it does not assume a parametric form for the displacement propagator. Second, its asymptotic behaviour is governed by a rigorous limit theorem (the parent paper's Theorem 1) tying the exponent of the persistence lifetime tail to the stability index of the underlying α-stable propagator. Third, the computation is $O(n_{\text{vox}} n_{\text{shells}} n_{\text{dirs}} \log n_{\text{dirs}})$ and parallelises trivially over voxels.

Limitations. The simulation calibration of Figures 4–5 reveals that, at HCP-like SNR (30) and shell schedule ($b \leq 3000$ s/mm^2), the per-direction angular variation in isotropic voxels is dominated by Rician noise rather than by the underlying displacement tail. The cumulative-sum construction (8) therefore produces a near-Gaussian bridge for moderate stability indices, and the Hill estimator (6) saturates near 2. The persistence-tail diagnostic provides its sharpest discrimination in two regimes: (i) simulated molecular trajectories (Figure 1), and (ii) strongly restricted environments (Figure 4 (c), where the lifetime count collapses to $\sim 10^2$). Improving the SNR-limited regime by combining direction subsampling, denoising [16], and longer b-shells ($b_{\max} \geq 6000$ s/mm^2, as in the MGH Adult Diffusion protocol) is the most promising route to a clinically useful $\widehat{\alpha}_{\text{persistence}}$ contrast.

Relationship to existing topology-based DW-MRI work. Previous topological-data-analysis applications to DW-MRI have focused on the persistence of *fibre-orientation distribution* fields [3, 17] or on connectomic graphs [12]. Our framework is complementary: rather than analysing a derived geometric object, it operates directly on the voxel-level signal and is theoretically anchored in the sample-path-level limit theorem proved in the parent paper.

Outlook. A natural next step is to combine $\widehat{\alpha}_{\mathrm{persistence}}$ with denoised, high-b-shell acquisitions and to evaluate clinical contrast in stroke (where ischaemic restriction sharply alters the displacement-tail structure) and tumour grading (where heterogeneous microenvironments produce mixed α-stable signatures). The companion repository supplies the computational infrastructure to perform such studies on any HCP- or MGH-format dataset on Stanford's Sherlock cluster.

Data and Code Availability

Code. The full source code, simulation pipeline, and SLURM submission scripts are released under the MIT licence at `https://github.com/csw-research/topological-dwmri`. The parent paper's repository (`https://github.com/csw-research/topological-levy-phase-transition`) is imported as a dependency without modification.

Data. The simulation study uses no human-subjects data; all results in Sections 4 are reproducible from the code alone. The in-vivo pipeline targets two public datasets: the Human Connectome Project S1200 release [15] (available from `db.humanconnectome.org` under the WU-Minn HCP open-access data use terms) and the MGH Adult Diffusion dataset [11] (available from the same portal). No personally identifiable information is processed or distributed by this work.

Reproducibility. All simulations are seeded; with the seeds and configurations recorded in `src/simulation/experi` every figure can be regenerated by executing `python scripts/run_all_experiments.py` followed by `python analysis/make_figures.py`.

Acknowledgements

We acknowledge the Wu-Minn Human Connectome Project (PIs D. Van Essen and K. Ugurbil; 1U54MH091657), funded by the 16 NIH Institutes and Centers that support the NIH Blueprint for Neuroscience Research, and the McDonnell Center for Systems Neuroscience at Washington University. Computation was performed on Stanford's Sherlock cluster.

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